

UGANDA CHRISTIAN UNIVERSITY

A Center of Excellence in the Heart of Africa

DEPARTMENT OF COMPUTING & TECHNOLOGY

Faculty of Engineering, Design & Technology

INDIVIDUAL CONTRIBUTION STATEMENTS

DSC2107 – Data Mining and Wrangling

Group 9

Web Scraping and Repeated Collection and Analysis of Allbirds Product Data

Reporting Period: 15 – 19 October 2026

Group Members: Agenorwot Cynthia Jillian
Ayoo Alice
Ngobi Jonah
Jjumba
Keji Flora

Mukono, Uganda

Individual Contribution Statement — Agenorwot Cynthia Jillian

Name: Agenorwot Cynthia Jillian
Course Unit: DSC2107 – Data Mining and Wrangling
Group: Group 9
Project: Web Scraping and Repeated Collection and Analysis of Allbirds Product Data
Reporting Period: 15 – 19 October 2026

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1. Overview of my role

I was one of the five students who carried out the core technical work of this project. I led the scraper development, the data cleaning and database pipeline, and the analysis notebooks, and I co-authored the final report.

2. Preliminary scraping practice (Day 1)

I wrote `day1_practice.ipynb`, which scraped *Books to Scrape* to practise extracting titles, prices, availability, star-ratings and product URLs and to follow pagination. The run collected 20 records before a read timeout; I used this failure as a design lesson that shaped the Allbirds scraper's retry and delay logic.

3. Website reconnaissance (Day 2)

I wrote `reconnaissance.ipynb`, which saved `robots.txt`, parsed it with `urllib.robotparser`, and confirmed that the `products.json` feed was permitted for our project User-Agent. I inspected a sample of the feed to confirm HTTP 200, `application/json`, and the product/variant field structure.

4. Scraper development (Day 3)

I developed `scraper.ipynb`: 250-item pages, deduplication by product ID, UTC-timestamped JSON snapshots that never overwrite earlier runs, 25 s timeout, 3–5 s random delays, four retries with exponential backoff, and hard stops on HTTP 401/403/429. Pagination was verified across three pages (250 + 250 + 192 = 692 products).

5. Repeated collection and data cleaning

I ran the scraper across five distinct dates (15–19 October 2026), producing seven raw JSON snapshots. I then wrote `clean_load.ipynb`, which cleans prices with regular expressions, identifies sizes from product options, standardises tags, removes duplicate observation keys, exports two cleaned CSVs, and loads 692 products and 52,178 observations into SQLite.

6. Database design

I designed the two-table schema in `data/allbirds.db`: `products` keyed on `product_id`, and `observations` with a foreign key and a `UNIQUE(product_id, variant_id, scraped_at)` constraint that allows safe reruns.

7. Analysis and visualisation

I wrote `analysis.ipynb` (six labelled charts) and `extra_analysis.ipynb` (daily availability summary, availability-percentage-by-day chart, and first-versus-last comparison showing 70 of 7,454 variants changed status). I also wrote `run_pipeline.ipynb` to orchestrate the steps.

8. Report writing

I co-authored the final report — title page, abstract, introduction, methodology, data management, ethical considerations, limitations, conclusion, references and appendices — and prepared the presentation slides.

9. Declaration

The work described above is my own. The other members of Group 9 contributed the additional tasks described in their own statements. Evidence of my contribution:

- `day1_practice.ipynb`, `reconnaissance.ipynb`, `scraper.ipynb`, `clean_load.ipynb`, `analysis.ipynb`, `extra_analysis.ipynb`, `run_pipeline.ipynb`
- `data/allbirds.db`, `data/clean/*.csv`, `data/raw/allbirds_raw_*.json`, `data/raw/robots_findi
- `outputs/charts/*.png`, `outputs/daily_summary.csv`
- `report.tex` / `WEB_SCRAPING.pdf`, presentation deck

Individual Contribution Statement — Ayoo Alice

Name: Ayoo Alice
Course Unit: DSC2107 – Data Mining and Wrangling
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1. Overview of my role

I worked alongside Cynthia to complete this project. My contribution focused on reviewing the cleaning and database logic, carrying out data validation and quality assessment, supporting the analysis, and co-authoring the report.

2. Preliminary practice and reconnaissance support

I contributed to the design of `day1_practice.ipynb` and reviewed its traceback after the read timeout, confirming the failure was a network timeout rather than a parsing error — which supported the decision to add retry logic to the Allbirds scraper. I also reviewed `reconnaissance.ipynb`, verifying that `robots.txt` was saved correctly, that the feed was permitted for our User-Agent, and that the response was valid JSON before scraping proceeded.

3. Data cleaning and database loading

I reviewed `clean_load.ipynb`: the price-cleaning function, the handling of missing or non-numeric product and variant IDs, the `available` flag normalisation, the tag standardisation, and the deduplication on `(product_id, variant_id, scraped_at)`. I also reviewed the SQLite loading logic — the upsert semantics for `products`, the insert-or-ignore semantics for `observations`, and the uniqueness constraint — and ran verification queries confirming 692 products and 52,178 observations.

4. Data validation

I carried out validation checks on the cleaned dataset: missing values per column (zero missing prices or identifiers), duplicate observation keys, price summary statistics (0.80–165, mean 101.84, median 105, IQR 75–130), availability counts (35,772 unavailable,

16,406 available), and product-type counts. I identified and documented the price minimum of 0.80 and the two non-standard product types (blank and `return,package_protection`), and proposed they be excluded from type-level averages. I also verified that every observation points to a valid product.

5. Analysis support

I reviewed the chart logic in `analysis.ipynb` and `extra_analysis.ipynb`, checking that each chart had a title and labelled axes. I reviewed `outputs/daily_summary.csv` and the first-versus-last comparison (70 of 7,454 variants changed status), and confirmed the availability-percentage-by-day chart was consistent with the daily summary figures (31.47, 31.74, 31.41, 31.23, 31.10).

6. Report writing and presentation

I co-authored the final report, concentrating on the methodology and data-management chapters, the data-quality assessment section, the ethical considerations and limitations, and the recommendations. I helped assemble the title page, references, appendices and checklist, and rehearsed the group presentation.

7. Declaration

The work described above is my own. The other members of Group 9 contributed the additional tasks described in their own statements. Evidence of my contribution:

- `day1_practice.ipynb`, `reconnaissance.ipynb`, `clean_load.ipynb`, `analysis.ipynb`, `extra_analysis.ipynb`
- `data/allbirds.db`, `data/clean/*.csv`, `outputs/daily_summary.csv`
- `outputs/charts/*.png`
- `report.tex` / `WEB_SCRAPING.pdf`, presentation deck

Individual Contribution Statement — Ngobi Jonah

Name: Ngobi Jonah
Course Unit: DSC2107 – Data Mining and Wrangling
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1. Overview of my role

I contributed to the written and presentational side of the project: the report introduction, the references section, the README documentation, and the presentation slides.

2. Literature and documentation review

I compiled and formatted the report's References section, verifying the Allbirds website, Python, Requests, Pandas, SQLite and Matplotlib documentation URLs cited in the final report. I drafted and maintained the `README.md`, documenting the project overview, objectives, technologies used, project structure, database description, installation steps, and execution order of the notebooks so that a reader can reproduce the pipeline from scratch.

3. Report introduction

I drafted the report's Introduction chapter — the background on e-commerce data and web scraping, the problem statement on messy product data and the need for a reproducible pipeline, and the scope statement. I ensured the introduction's objectives align with the actual deliverables produced by the technical team.

4. Presentation preparation

I contributed to the presentation deck: planning the slide flow (problem, objectives, tools, methodology, results, charts, findings and recommendations), drafting speaker notes, and rehearsing the delivery. I also prepared the charts from `outputs/charts/` for the slides in the correct order (Figures 1–6, plus the availability-percentage-by-day chart).

5. Verification and review

I reviewed the final report for consistency with the actual results — product count 692, observation count 52,178, five distinct collection dates, availability around 31.5 %, and 70 changed variants — and checked that the ethical-considerations and limitations sections reflect the actual project limitations (the practice-run timeout, the 5 vs 7 days, and the two non-standard product types).

6. Declaration

The work described above is my own. The other members of Group 9 contributed the additional tasks described in their own statements. Evidence of my contribution:

- `README.md`
- `report.tex` (Introduction chapter, References section)
- presentation deck (slides and speaker notes)
- review notes on the final report's consistency with the actual results

Individual Contribution Statement — Jjumba

Name: Jjumba
Course Unit: DSC2107 – Data Mining and Wrangling
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1. Overview of my role

I contributed to the report's front matter, appendices and submission package, and supported the group's project organisation and documentation.

2. Title page and front matter

I prepared and typeset the report's title page — UCU logo, university and department headings, project title, group information table (course unit, programme, reporting period, group members, lecturer, submission date) — and fixed the initial compile errors in the title page (the missing `tabular` environment and the quoted logo filename with a space). I also assembled the table of contents, list of figures and list of tables.

3. Appendices

I compiled the Appendices chapter: the group contributions table and the final submission checklist. I ensured every checklist item (collection dates, valid snapshot count, product and observation counts, validation table, charts and interpretation, temporal findings, failed-request documentation, group contributions, notebook and database verification) was either completed or explicitly flagged.

4. Submission packaging

I organised the final submission package: verified that the notebooks run in the correct order (`day1_practice`, `reconnaissance`, `scraper`, `clean_load`, `analysis`, `extra_analysis`), that `data/allbirds.db` and the cleaned CSVs are present, and that all chart outputs in `outputs/charts/` exist and are legible. I compressed the raw JSON snapshots and excluded the virtual environment and notebook caches so the submission stays within the size limit.

5. Review

I reviewed the report for typographical and formatting consistency: heading levels, table alignment, figure captions, and the correct rendering of numbers (e.g. 52,178 and 7,454 rather than 52,178/7,454 as text).

6. Declaration

The work described above is my own. The other members of Group 9 contributed the additional tasks described in their own statements. Evidence of my contribution:

- `report.tex` (title page, front matter, Appendices A and B)
- submission package (`Group9_Submission.zip`)
- formatting and consistency review notes

Individual Contribution Statement — Keji Flora

Name: Keji Flora
Course Unit: DSC2107 – Data Mining and Wrangling
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1. Overview of my role

I contributed to quality control on the database and charts, drafted the report's ethical-considerations and limitations sections, and carried out the final proofreading pass on the report and presentation.

2. Database schema review

I reviewed the SQLite schema in `data/allbirds.db` against the report's database-design chapter: the `products` table (primary key, fields) and the `observations` table (foreign key, `UNIQUE(product_id, variant_id, scraped_at)` constraint). I confirmed the field names in the database match those documented in the report and that the foreign-key relationship is enforced (`PRAGMA foreign_keys = ON`).

3. Chart quality control

I checked every chart in `outputs/charts/` for completeness: each has a title, labelled axes with units ("Price (source currency)", "Number of observations", "Observation count"), and readable tick labels. I flagged that the average-price-by-type chart includes blank and `return,package_protection` categories that should be interpreted with caution, and ensured the report states this.

4. Ethical considerations and limitations

I drafted the report's Ethical Considerations and Limitations sections. The ethical section documents that the group checked `robots.txt`, honoured the applicable restrictions, used controlled request frequency (3–5 s delays), and stopped on refused access. The limitations section documents that only five distinct calendar dates yielded valid snapshots (not seven), that the Books to Scrape practice run timed out before writing its CSV, that

the price minimum of 0.80 and the `return,package_protection` product type suggest non-standard records, and that the results are descriptive only.

5. Final proofreading

I proofread the final report and presentation for consistency with the actual results: 692 products, 52,178 observations, 5 distinct dates, mean price 101.84, median 105, 31.5 % availability, and 70 of 7,454 variants changed. I also checked that no placeholder text (e.g. "[Insert lecturer's name]", "[Insert date]") remained in the submitted report.

6. Declaration

The work described above is my own. The other members of Group 9 contributed the additional tasks described in their own statements. Evidence of my contribution:

- `report.tex` (Ethical Considerations and Limitations sections)
- chart quality-control notes on `outputs/charts/*.png`
- database schema review notes on `data/allbirds.db`
- final proofreading notes on the report and presentation